Kept

A guarded next best action for promises customers can keep.

Team Sharpe Shooters · CIBC Collections Hackathon · 4 Oct 2026

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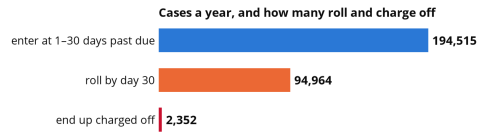
1.05M
people, from 3.84M source keys

96,000
guarded actions a night

62% vs 52%
of promised money paid: payday-dated vs not

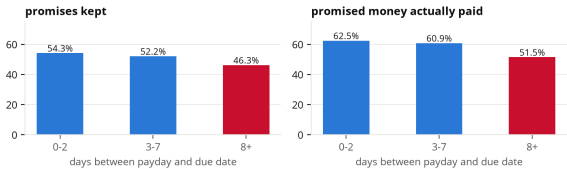
The problem: generic treatment, promises set up to break

- **One person, five IDs.** CRM, cards, loans, deposits and collections each key customers their own way. Agents see a slice.
- **The facts are in free text.** “Lost shifts at the plant, can pay 50 on the 15th” lives in a note no system reads.
- **Promises fall due before payday.** Same reminder, same day, for everyone. They break; the account rolls.
- **Hardship found late.** Pressure on people who are already struggling, and regulator risk for the bank.



48.8% of cases entering at 1–30 days past due roll by day 30. Rolling is the *only* road to a charge-off: 2.5% of rolled cases charge off (avg $12.5K); 0% of the rest.

The insight: when you ask matters as much as who you ask



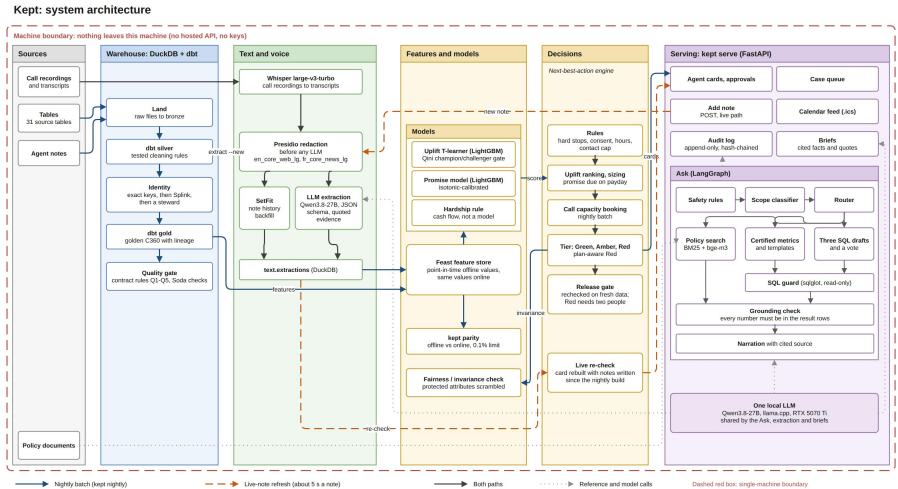
32,000 resolved promises where we can see the customer's salary deposits (bank's own history).

Kept = three ideas, in this order

1. **Safety first.** Hard stops, consent, calling hours, contact caps and hardship are rules, not model outputs.
2. **Uplift, not "likely to pay".** Contact the people the action actually helps.
3. **Promises they can keep.** Due 1–2 days after payday, sized to spare cash.

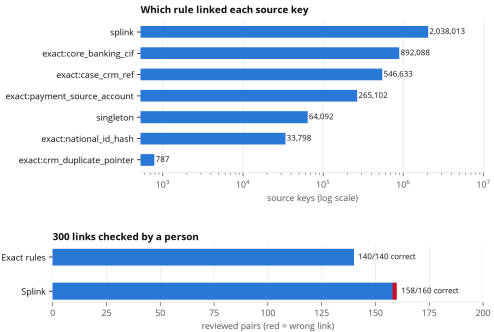
A person releases every action.

The system: four layers, governance through all of them



Full size: `docs/architecture/architecture.pdf`

Layer 1 · one customer, one ID, a contract that blocks bad data



3.84M → 1.05M

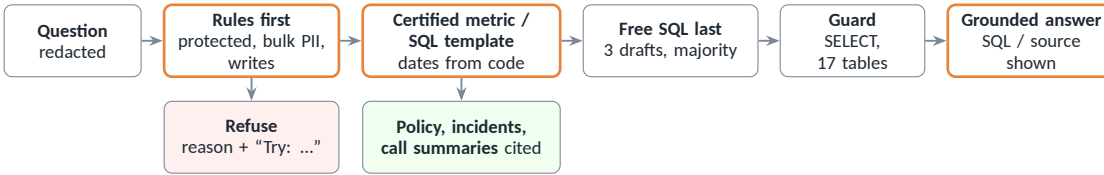
source keys → golden IDs

99.3%

match precision, 300 links checked by a person

- Exact keys, then Splink: ≥0.95 auto, 0.80–0.95 to a steward
- **19 data traps** counted and handled; never guess
- Contract from DC-COLL-001; gate **PASS**, all 8 rules

Layer 2 · plain-English questions, answers with their working



94%
dev accuracy (target 80%)

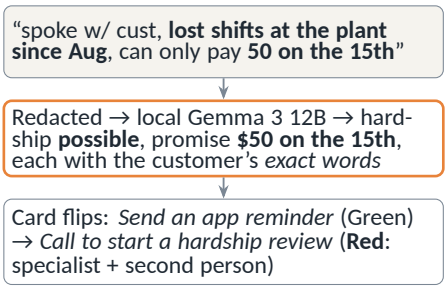
100%
refusal recall (target 95%)

100%
refusal precision

Every number in an answer must appear in the result rows. Protected-attribute questions are refused *before* any LLM sees them; close-sounding ones (“agents certified for vulnerable customers”) still get answered.

Layer 3 · a note changes the action, in about five seconds

Live, on real data:



0 leaks

in 785,594 redacted notes and calls

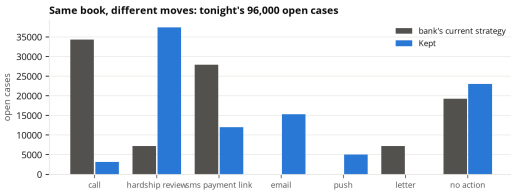
0.98 / 0.99

hardship recall, notes / calls (held out)

- No quote, no value; text cease needs an agent
- SetFit backfills 785K old texts in minutes
- Cash-flow rule: late pay, pay cut, new EI
- Feast: train = live (0 of 56,000 off)

Layer 4 · the guarded next best action



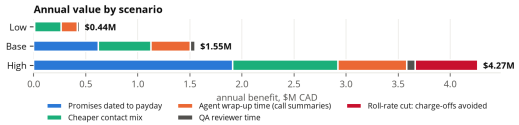


+21 pts

right-party contact for evening calls vs daytime calls (uplift study); contact itself barely moves payment

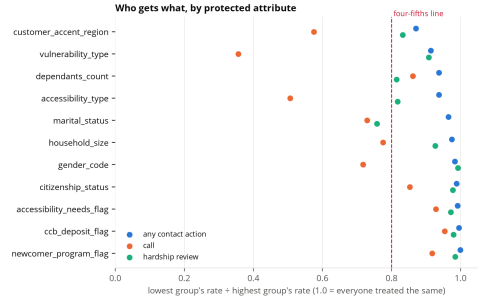
Same number of calls as today, but 37K become **hardship reviews**; $1.10 letters become near-free texts. Every card: three plain-English reasons, similar past cases, the checks that passed. Released by a person, re-checked at release.

Does it work, is it fair, what's it worth?



$2.2M/yr
base case ($1.0–4.2M)

3.8×
year-1 ROI, payback ~2 mo



Fair: any-contact rate within 0.86–1.00 across every protected group; flags only on rare calls and protective reviews.

Measured: a 5% hold-out stays on the old strategy; a kill switch reverts all.

Built responsibly, built together, and what's next

Responsible by construction

- Human releases every action
- Rules before and after the model
- No protected attributes, tested
- Redaction before any model
- Free / local LLMs, all audited

Built together

- 4 tracks: data · Ask · text · model + UI
- Typed contracts between tracks, fakes for each
- 5 integration gates (G0–G4)
- 64 PRs · 315 commits · 1,200+ tests
- One nightly: ~7 min, 1M people

Next

- Retrain with payroll features (drift report found the gap)
- Plan-aware rule to cut Red reviews
- Score every call for QA with a fine-tuned fast classifier
- Read the hold-out at day 30

Promises customers can keep.